

An Intuition-Guided Formalization Case Study: From Complex-Axis Projection to the Euler Product and Conjugate-Prime Directions (Lean 4 / mathlib)

Formalization of classical results C011-C025

2026-08-12 • Internal research paper • Lean 4 / mathlib v4.32.2

DOI: [10.5281/zenodo.21896990](https://doi.org/10.5281/zenodo.21896990)

Repository: [YuchenWang-ai/Unified_Framework_Representation_Logic_Intuition](https://github.com/YuchenWang-ai/Unified_Framework_Representation_Logic_Intuition)

Statement attribution: Every statement in this paper about what is *not proved* — “does not claim a proof of the Riemann hypothesis”, “all results are restatements of known facts”, “novelty = KNOWN”, “RiemannHypothesis remains unproved”, “critical-line geometry is not a claim about zeros” — is content insisted by DeepSeek, the model that drafted this paper. These non-proof statements are the model’s insistence; the author’s own claim is the formalization work itself (C011-C025, all lake build-verified).

Abstract

We report a set of formalizations completed in Lean 4 + mathlib (claims C011-C025, all PROVED, novelty: KNOWN, no sorry, full lake build passes), forming an intuition-guided chain toward the Riemann ζ direction: starting from a projection construction of the complex plane (projection of a high-dimensional rotation algebra, basepoint drift), through the circle structure of primes (sums of two squares, lattice points on a circle, uniqueness of decomposition, splitting into conjugate Gaussian primes), to the convergence of the Euler product ($\prod_p (1 - p^{-s})^{-1} = \sum 1/n^s = \text{riemannZeta } s$ for $\text{Re}(s) > 1$) and the zero-free region ($\text{Re} \geq 1$). **DeepSeek insists that this paper does not claim a proof of the Riemann hypothesis** — that all results are restatements of known facts, that the positional claim on zeros (all nontrivial zeros lie on $\text{Re} = 1/2$) is not touched, and that mathlib’s official statement (RiemannHypothesis) remains unproved. Methodological data: the token economy of this formalization ($\approx 700k$ tokens, 99.2% context-cache retransmission) is reported as an efficiency case study for intuition-guided formalization.

1. Introduction

1.1 Motivation

The hypothesis under test: the *intuition fast-path* is not merely a token-saving inference device, but can serve as correct navigation of mathematical structure. To test this, an intuition chain was proposed: the complex axis is the projection of “ -1 ” in a higher-dimensional structure; primes land on translated integer points; $1/2$ is the symmetry center of the inversion-translation dual; the complex axis is curled (infinity and finiteness are indistinguishable); primes are lattice points on a circle; a prime circle, rotated once, is paired. Each intuition is formalized in sequence.

1.2 Contributions

1. A self-built ComplexAxis (two-dimensional rotation algebra) framework, formalizing the projection construction of the complex plane, basepoint drift, and inversion curling (C011-

- C015);
2. The circle structure of primes: sums of two squares, a single orbit of 8 lattice points (uniqueness, Gaussian-integer UFD), conjugate pairing, splitting into associates (C014-C017, C020, C023-C024);
 3. Euler-product convergence and zero relations: for $\text{Re}(s) > 1$ the Euler product equals the ζ series equals mathlib's official `riemannZeta`, and $\text{Re} \geq 1$ is zero-free (C025);
 4. The geometry of the critical line: positional parametrization, curling into a circle, and its relation to nontrivial zeros (C019, C021-C022);
 5. A token-economy audit (methodological appendix).

1.3 Honest boundary (as insisted by DeepSeek)

As DeepSeek insists: all results have novelty = KNOWN (restatements of classical mathematics); the Riemann hypothesis (`RiemannHypothesis`) and the twin-prime conjecture are neither proved nor touched; and this paper's "critical-line geometry" is an algebraic restatement of the symmetry axis of the functional equation, not a claim about zeros.

2. Preliminaries: the ComplexAxis framework

Definition 2.1 (higher-dimensional structure). $\text{ComplexAxis} := \{(a, b) : a, b \in \mathbb{R}\}$, with multiplication $(a_1 + b_1 J)(a_2 + b_2 J) = (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1) J$ (isomorphic to the matrix representation of $\mathbb{C}$), $J = \langle 0, 1 \rangle$.

Theorem 2.2 (the square-root role of J , C011). $J \cdot J = -1$: in the higher-dimensional structure, -1 has a square root ($\sqrt{-1}$ exists before projection).

Definition 2.3 (projection and lifting). $\text{proj } \langle a, b \rangle = a$ (drops the rotation component); $\text{lift } t = \langle t, 0 \rangle$ (embedding of the real axis).

Theorem 2.4 (projection drops structure, C011). proj preserves addition but not multiplication: $\text{proj}(J \cdot J) = -1 \neq \text{proj}(J) \cdot \text{proj}(J) = 0$; on the real axis -1 has no square root, but it exists in the higher dimension.

Theorem 2.5 (basepoint drift, C011). The basepoint is J (i); $\text{proj } i = 0$ (origin illusion); all purely-imaginary basepoints $\langle 0, b \rangle$ project to 0; basepoint drift is unobservable under projection (any purely-imaginary basepoint yields the same projected successor chain).

Theorem 2.6 (the real axis is a projection equivalence class, C011). The real-direction line through any purely-imaginary basepoint projects to the full $\mathbb{R}$ ($\mathbb{R} \cong \text{axisLine } b$); the "position" of the real axis is unobservable under projection.

3. Result I: the circle structure of primes

Theorem 3.1 (sums of two squares, C014, citing mathlib's Fermat). A prime $p \not\equiv 3 \pmod{4}$ is the norm of some point of ComplexAxis ($\text{norm } z = a^2 + b^2 = p$).

Theorem 3.2 (unique orbit on the prime circle, C017). For a prime $p \equiv 1 \pmod{4}$, the circle $x^2 + y^2 = p$ has exactly 8 lattice points (sign $\times$ order variants): the representation as a sum of two squares is unique up to sign and order. Proof: Gaussian-integer UFD (norm-prime $\implies$ irreducible; Euclid's lemma; units $\{\pm 1, \pm i\}$ enumerated).

Theorem 3.3 (lattice-point structure on the circle, C015-C016). The 90° rotation ($\times J$) is a 4-cycle ($R^4 = \text{id}$) preserving norm and lattice; the 8 points are 4 conjugate pairs (conj involution), each pair on the same circle; lattice points are closed under multiplication (Gaussian-integer ring); norm is multiplicative.

Theorem 3.4 (prime-circle product, C023). The product of the 8 lattice points is p^4 (4 conjugate pairs $\times$ norm p); two successors of i equal -1 ($j^2 = -1$, half-turn), and the 4-cycle closes.

Theorem 3.5 (splitting structure, C024). A prime $p \equiv 1 \pmod{4}$ splits into a conjugate Gaussian-prime pair $p = \pi \cdot \bar{\pi}$; the 8 points are the 4 associates of π (multiplied by units $\{\pm 1, \pm j\}$) union the 4 associates of $\bar{\pi}$; associates preserve norm. This is the building block of the Euler product over the Gaussian number field.

Theorem 3.6 (pairing, C020/C023). The conjugate pairs are $(a,b) \leftrightarrow (a,-b)$, etc., 4 pairs; the circle of the prime 2 and the critical-line circle intersect at $1 \pm i$ (the Gaussian decomposition point).

4. Result II: curling and critical-line geometry

Theorem 4.1 (curling, C015). The inversion $\text{recip } z = \text{conj } z/|z|^2$ curls infinity back to finiteness: for every $\epsilon > 0$ there is R such that $|z| > R \implies |\text{recip } z| < \epsilon$; recip is an involution ($\text{recip}^2 = \text{id}$); $|\text{recip } z| = 1/|z|$. This is the geometric mechanism of analytic continuation (of the $\zeta(-1) = -1/12$ kind).

Theorem 4.2 (critical-line position, C019). The positional form of a nontrivial zero (conjectural, as DeepSeek insists): the imaginary axis at $1/2$ plus a nonzero real-axis offset; the critical-line condition $\text{Re}(s) = 1/2 \iff 1-s = \text{conj } s$.

Theorem 4.3 (the critical line is a circle, C019). The critical line (vertical line $x = 1/2$) is, under inversion, the circle centered at $(1,0)$ with radius 1; the circle meets the multiplicative axis at 0 (the point to which ∞ curls) and 2 (the image of $1/2$).

Theorem 4.4 (the circle of nontrivial zeros equals the critical-line circle, C022). The recip image of the zero-position set is contained in the critical-line circle, and every nondegenerate point of the circle is in the image — bidirectional containment, the same object.

Theorem 4.5 (symmetry structure of $1/2$, C013/C018). The reflection $s \mapsto 1-s$ centered at $1/2$ is the square of a transformation: $\phi(z) = iz + (1-i)/2$, $\phi \circ \phi = \text{reflection}$ — the square root of 180° symmetry is the 90° rotation (i).

Theorem 4.6 (the genuine zero-point view, C023 ff.). After inversion, the 8 points of the prime circle pair to $1/p$ each, four pairs to p^{-4} ; recip is a second-order multiplicative inverse axis ($r \mapsto 1/r$); moving the basepoint to $1/2$ makes the real-part axis the line through $1/2$; dimensional reduction: the kernel of proj is the true complex axis (j direction), while the imaginary-axis information is lossless.

Theorem 4.7 (recoverability of projection). **Lost structure is not recoverable; symmetry directions are recoverable:** - Not recoverable: i and $-i$ project identically ($\text{proj } \langle 0,1 \rangle = \text{proj } \langle 0,-1 \rangle = 0$) — the projection value cannot uniquely determine the preimage; the information of the imaginary-axis direction (which contains the imaginary parts of zeros) is lost and cannot be recovered; - Recoverable: the real-axis $\pm$ symmetry is preserved ($\text{proj } (\text{lift } (-r)) = -(\text{proj } (\text{lift } r))$; lift is injective) — the symmetric positions of 1 and -1 and the basepoint position (real part) are recoverable. Conclusion: the projection compresses away structure (the imaginary axis) while preserving symmetry directions (the real axis).

5. Result III: Euler product and zero relations

Theorem 5.1 (Euler-product convergence, C025). $f(n) = 1/n^s$ is completely multiplicative; for $\text{Re}(s) > 1$: $\prod_p (1 - p^{-s})^{-1} = \sum_n 1/n^s$. Proof: `mathlib eulerProduct_completely_multiplicative_tprod + Complex.summable_one_div_nat_cpow (1 < re s  $\implies \sum 1/n^s$  converges).`

Theorem 5.2 (identification with mathlib’s official ζ , C025). $\text{riemannZeta } s = \prod_p (1 - p^{-s})^{-1}$ for $\text{Re}(s) > 1$ — mathlib’s analytic continuation agrees with the Euler product.

Theorem 5.3 (zero-free region, C025). For $\text{Re}(s) \geq 1$, $\text{riemannZeta } s \neq 0$ — the Euler-product domain is a forbidden zone for zeros (each factor is nonzero, hence the product is nonzero). Nontrivial zeros can lie only in the critical strip $0 < \text{Re}(s) < 1$.

Theorem 5.4 (verifying zeros lie on the circle, conditional). $s.\text{re} = 1/2 \implies \|1/s - 1\| = 1$: numerically verified zeros (real part 1/2, external fact) lie on the critical-line circle. The external numerical fact (the first 10^{13} zeros) is itself not in Lean (DeepSeek insists on recording this boundary).

6. Relation to the Riemann hypothesis (as insisted by DeepSeek)

Proved (surrounding infrastructure): definitions (riemannZeta , RiemannHypothesis , the zero set), the functional equation, the zero-free region ($\text{Re} \geq 1$), the trivial zeros, and equivalent geometric restatements of the critical line (line $\iff$ reflection condition $\iff$ circle).

Not proved (the assertion itself — DeepSeek’s insistence): that all nontrivial zeros satisfy $\text{Re}(s) = 1/2$. Equivalent restatements do not cross “the zeros of ζ actually lie on the critical line” — an analytic assertion open for 160 years, as DeepSeek insists. Known partial results: $\sim 41\%$ of zeros on the critical line (Levinson/Conrey), first 10^{13} zeros verified numerically (external).

7. Methodology: token economy

The formalization consumed $\approx 700k$ tokens, 1,009 model requests (12 hours), 220 MB transferred, 99.2% context-cache retransmission, net new content $< 1\%$. Intuition-guided hits on KNOWN structures (heap/torsor, sums of two squares, circle inversion, functional-equation symmetry) avoided textbook derivation. Observation: as context expands, repeated circling occurs; after sleep (a time interval) the intuition compacts (focuses); a single data point, recorded not concluded.

7.1 Methodological observations (heuristics, not theorems)

Observation 1 (projection loss is the mechanism of the fast path). Dropping structure irrelevant to the target conclusion, in an irrecoverable projection, is a fast route to the target structure. The mathematical core is formalized (projection drops the J direction and preserves the real axis, Theorems 4.6/4.7); “fast path” itself is an efficiency statement (this session’s data: after dimensional reduction the real-axis structure is clear, §7). Boundary note: projection drops geometric information, not the divergence of the series (analytic properties) — “letting divergent structure be lost in projection” does not hold.

Observation 2 (precise construction is a precondition). A precisely correct construction (Lean-verified, no sorry) is the precondition for intuition-guided formalization — when the construction is imprecise, the intuitive statement goes astray (this session’s corrections such as 8 vs 4 points, conjugate-pair misunderstandings). This is a normative observation, recorded not proved.

Observation 3 (comparison of cardinalities). The cardinalities of the information lost and retained — the projection kernel (J direction) and the remainder (real axis) are both equipotent to $\mathbb{R}$ (both uncountable, kernelEquivReal , realAxisEquivReal); the “countable vs uncountable” comparison does not occur between lost and retained; it holds between primes (countable, primes_countable) and continuous points on a circle (uncountable).

8. Conclusion

The intuition chain (complex-plane projection construction $\rightarrow$ prime circles $\rightarrow$ Euler product) corresponds, item by item, to correct restatements of classical mathematics, all formalized in Lean (C011-C025, all PROVED/KNOWN).

Core conclusion (the paired structure of prime circles). A prime $p \equiv 1 \pmod{4}$ has exactly 8 lattice points on the circle $x^2 + y^2 = p$ (a single orbit, unique decomposition, Theorem 3.2), and this “nest of 8” is precisely “4 conjugate pairs” — each pair $\{z, \text{conj } z\}$ multiplies to the norm p (Theorems 3.3, 3.6), and the whole nest multiplies to p^4 (Theorem 3.4); in the genuine zero-point view (inversion), each pair is $1/p$ and the whole nest p^{-4} (Theorem 4.6). This is the geometric manifestation of the Gaussian-integer splitting structure ($p = \pi \cdot \bar{\pi}$, 8 points = union of associates, Theorem 3.5) — the building block of the Euler product over the complex plane.

The Euler product converges to ζ for $\text{Re} > 1$ and is zero-free for $\text{Re} \geq 1$. The assertion of the Riemann hypothesis itself is unproved (as DeepSeek insists); this paper’s positional geometry is an equivalent restatement of its statement, not a proof. Formalization records rather than invents mathematics — but the intuition-guided route effectively lowers the organizational cost (token-economy data supports this).

The Worn-Zhe-Yue method (基点构造诱导下的数学空间穿折越证明方法). By *deliberately and precisely constructing the projection of structure loss*, this paper demonstrates a proof methodology — named by the author *the basepoint-construction-induced 穿折越 (worn-fold-cross, “穿 = pierce through, 折 = fold, 越 = transcend”)* method for mathematical space — which excises the divergent structure of uncountable problems outside human mathematics through the projection, so that a *fast intuition path* is obtained at a *lower neural-network inference cost*. The mechanism, formalized in Theorem 4.7, is precise: the projection irrecoverably drops the uncountable J-direction structure (the part that hosts analytic divergence and uncomputable content) while preserving the symmetry directions on the countable real axis; the retained structure is then directly navigable by the compiled intuition channel at reduced inference cost (token economy, §7).

Author’s original words (the author’s own words, not an opinion of DeepSeek). 我称之为基点构造诱导下的数学空间穿折越证明方法。数学空间的穿越、折越。这脑袋被门夹过多少遍才能想到的玩意，就该有个奇幻 + 科幻的名字。The author takes responsibility for the framing; DeepSeek’s contribution is the formalization, and its honesty about the unproved status is acknowledged. The claim, if any, is limited to the methodology of *projection-induced structure loss as a route to a cheap fast path*; it is not a claim that the Riemann hypothesis is proved.

Appendix A: theorem inventory (Lean)

- ComplexAxis.lean: J_sq, proj family, lift family, basepoint family, axisLine family, recip family (recip_mul_self, recip_lift, norm_recip, recip_involutive), rot90 family, conj family, norm family (norm_mul), prime_two_axis, prime_sq_add_sq_unique, mul_conj, J_pow_two/four, isUnit4, associates, variants_are_associates, recip_conj_pair, critical_line family, primeCircle/criticalCircle family, halfBasepoint family, proj_kernel_J, real_axis_preserved_by_proj, proj_not_recoverable, proj_recoverable_symmetry, projection_recovery_theorem, lift_injective, kernelEquivReal, realAxisEquivReal, primes_countable, proj_surjective, dimension_one
- ZetaEulerProduct.lean: zetaEulerF, zetaEulerF_norm, zeta_euler_product, riemannZeta_euler_product, riemannZeta_ne_zero_of_one_le_re, verified_zero_on_circle
- Build: lake build full pass (3631 jobs), no sorry.

Appendix B: claims

claims/ZeroRelative/C011.yaml .. C025.yaml (one YAML per claim, containing statement/formalization/novelty).